

AI-Powered Enterprise Assistant

Intelligent Document Analysis,
RAG,
SQL Analytics &
Automated Reporting

Doc video link:

https://drive.google.com/file/d/1_6DnImI9M0kE9TODdB_jgQlay270UFxg/view?usp=drive_link

Problem Statement

Why an enterprise needs one intelligent information layer

- Enterprise information is distributed across documents, PDFs, scanned files, business databases, reports and dashboards.
- Employees spend time searching, extracting information and converting business questions into SQL.
- Goal: build one AI platform that can understand documents, answer questions, analyze structured data and generate reports.

Proposed Solution

AI Enterprise Assistant — one platform for documents and data

Document Intelligence

Upload and process enterprise PDFs and documents.

OCR + RAG

Extract text from scans and retrieve relevant context.

Natural-Language SQL

Ask business questions without writing SQL.

Analytics Dashboard

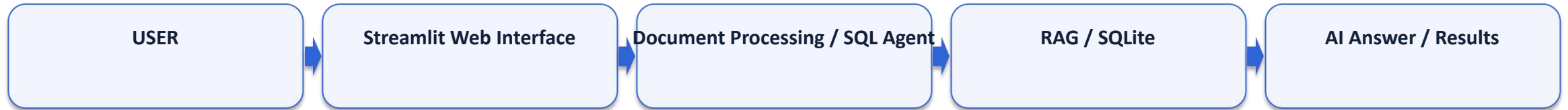
Visualize sales and business performance.

AI Report Generation

Turn analysis into executive-ready reports.

System Architecture

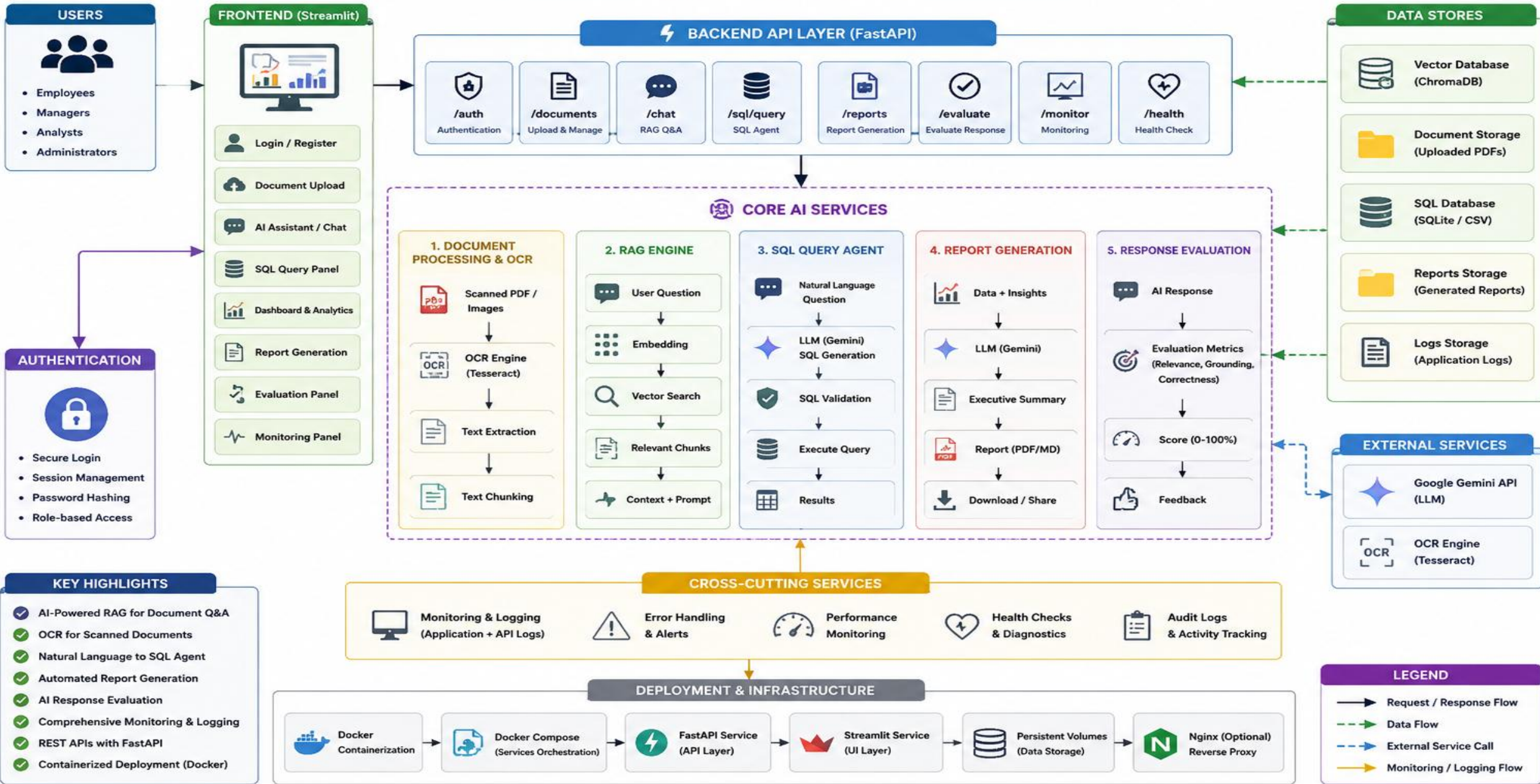
End-to-end flow from user request to grounded answer



- Documents: PDF → OCR → Text → Chunking → Embeddings → RAG
- Structured data: Natural language → SQL generation → Read-only SELECT → Result

AI-POWERED ENTERPRISE ASSISTANT

System Architecture



Document Processing & OCR

Making scanned and digital PDFs searchable

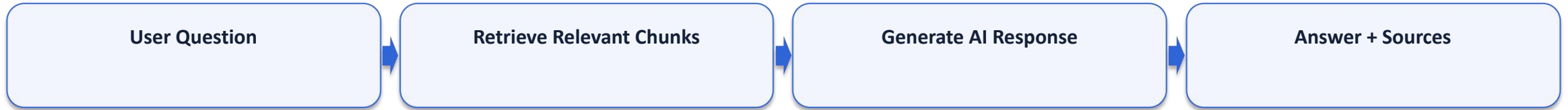
- PDF upload and text extraction
- Scanned-PDF detection
- Tesseract OCR for image-based pages
- Text chunking for retrieval
- Source-aware retrieval for grounded answers

Demo evidence

Scanned leave-policy PDF successfully processed and made available for question answering.

RAG Question Answering

Retrieve the right context before generating an answer



Example question

“Who approves planned leave requests?”

Example answer

“Planned leave requests must be submitted to the reporting manager for approval.”

Enterprise AI

Logged in: admin

- Navigation
- ☐ Dashboard
 - ☒ Documents & RAG
 - ☐ SQL Agent
 - ☐ AI Reports
 - ☐ Evaluation
 - ☐ Monitoring

Logout

Share ☆ ✎ ↺ ⋮



Document Intelligence & RAG

Upload enterprise PDF

 scanned...cy.pdf.pdf 302.4KB +

Processed scanned_leave_policy.pdf.pdf — 3 chunks indexed.

RAG is ready. Ask a question below.

Ask about your uploaded document

Who approves planned leave requests?

Ask AI

🗣️ Answer

Planned leave requests must be submitted to the reporting manager for approval. (ENTERPRISE EMPLOYEE LEAVE POLICY, Page 1)

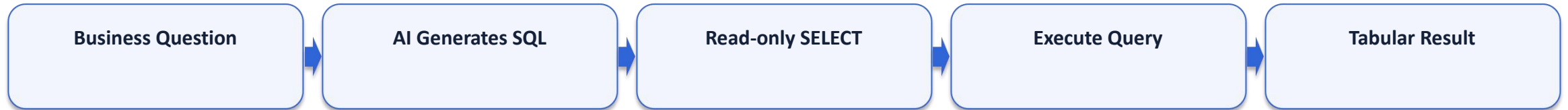
Source document: scanned_leave_policy.pdf.pdf

 Retrieved Sources

< Manage app

SQL Agent

Natural-language access to structured business data



Example

“What were total sales in July?” → AI generates SQL → executes query → displays the result.

Enterprise AI

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Logout

📄

Natural Language SQL Agent

	id	product	department	amount	month
0	1	Laptop	IT	50000	July
1	2	Phone	Sales	30000	July
2	3	Monitor	IT	25000	August
3	4	Laptop	Sales	60000	August
4	5	Tablet	Sales	40000	September
5	6	Monitor	IT	35000	September

Ask a database question

“What were the total sales in July?”

Run SQL Agent

```
SELECT SUM(amount) AS total_sales FROM sales WHERE month='July';
```

Query executed successfully.

	total_sales

< Manage app

Enterprise AI

Logged in: admin

- Navigation
- ☒ Dashboard
 - ☐ Documents & RAG
 - ☐ SQL Agent
 - ☐ AI Reports
 - ☐ Evaluation
 - ☐ Monitoring

Logout



Enterprise Command Center

Documents
0

AI Queries
0

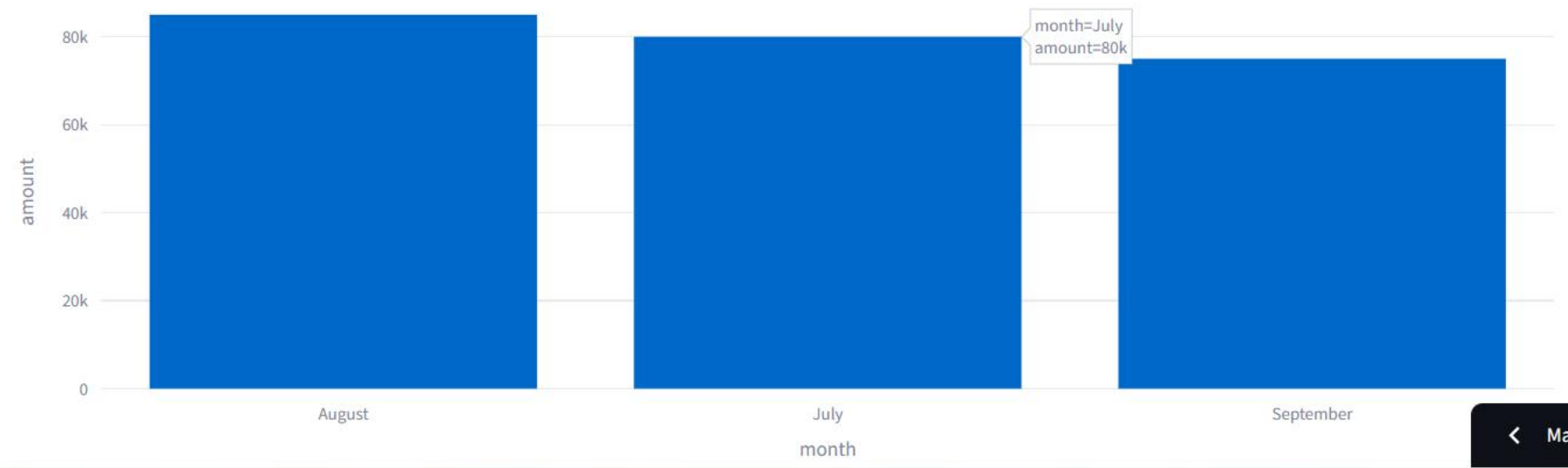
System Status
Online

RAG Index
Waiting

Sales Analytics



Monthly Sales



< Manage app

Dashboard & Reports

From raw business data to decision-ready insights

Sales Analytics

Monthly sales trends and performance indicators.

Data Table

Queryable results in a structured tabular view.

Executive Report

AI-assisted summary and downloadable report.

AI Evaluation

Measure whether generated answers satisfy expected criteria

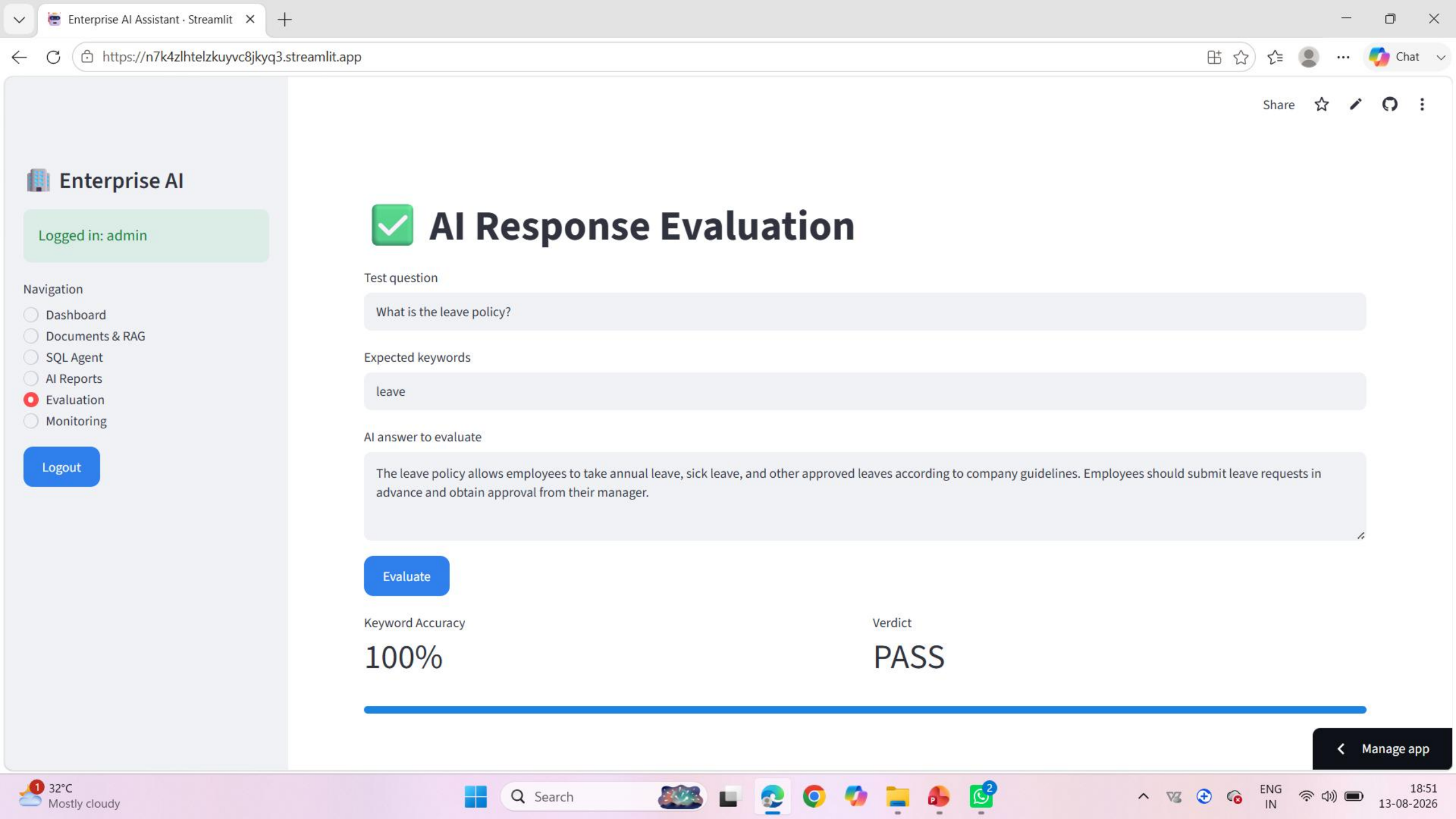
- Test question
- Expected keywords
- AI-generated answer
- Keyword accuracy
- Pass / Fail verdict

Example evaluation

Keyword Accuracy: 100%

Verdict: Pass

Replace with the actual evaluation result from the application if available.



- Navigation
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 - ☐ AI Reports
 - ☒ Evaluation
 - ☐ Monitoring

Logout

AI Response Evaluation

Test question

What is the leave policy?

Expected keywords

leave

AI answer to evaluate

The leave policy allows employees to take annual leave, sick leave, and other approved leaves according to company guidelines. Employees should submit leave requests in advance and obtain approval from their manager.

Evaluate

Keyword Accuracy

100%

Verdict

PASS

Manage app

Monitoring & APIs

Operational visibility and programmatic access

Monitoring

Application logs
Query latency
Logged events
Health status

REST APIs

/health
/documents/upload
/sql/query
/chat

Swagger / OpenAPI documentation

Enterprise AI Assistant · Streamlit

+

https://n7k4zlhtelzkuyvc8jkyq3.streamlit.app

Share ☆ ↺ ⋮

Enterprise AI

Logged in: admin

Navigation

- Dashboard
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- SQL Agent
- AI Reports
- Evaluation
- Monitoring

Logout

2026-08-13 11:57:22,162 | INFO | RAG query | user=admin | latency=1.58

2026-08-13 11:57:23,352 | INFO | HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash

2026-08-13 11:57:50,730 | INFO | AFC is enabled with max remote calls: 10.

2026-08-13 11:57:52,825 | INFO | HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash

2026-08-13 11:58:16,447 | INFO | AFC is enabled with max remote calls: 10.

2026-08-13 11:58:31,054 | INFO | HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash

2026-08-13 13:18:22,021 | INFO | AFC is enabled with max remote calls: 10.

2026-08-13 13:18:47,753 | INFO | AFC is enabled with max remote calls: 10.

2026-08-13 13:19:05,325 | INFO | HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash

2026-08-13 13:19:14,055 | INFO | HTTP Request: POST https://generativelanguage.googleapis.com/v1beta/models/gemini-2.5-flash

Health

API/UI service: Healthy

Database: Healthy

RAG engine: Ready

32°C Mostly cloudy

Search

Windows Taskbar

System Tray

18:51 13-08-2026

Manage app

Deployment & Technology Stack

Built with practical, production-oriented components

- Python • Streamlit • FastAPI • SQLite
- Tesseract OCR • RAG / Embeddings
- Pandas • Plotly
- GitHub • Streamlit Cloud

Deployment

Cloud deployment through Streamlit Cloud.

Actual app URL:

<https://n7k4zlhtelzkuyvc8jkyq3.streamlit.app/>

Results & Conclusion

- ✓ Enterprise documents can be uploaded and analyzed
- ✓ Scanned PDFs processed using OCR
- ✓ Context-aware RAG answers with sources
- ✓ Natural-language SQL analytics
- ✓ Automated executive reports
- ✓ Evaluation and monitoring included
- ✓ Cloud deployment available

Thank You